

11.1 Dipole Radiation

11.1.1 What Is Radiation?

When charges *accelerate*, their fields can transport energy irreversibly out to infinity – a process we call **radiation**.¹ Let us assume the source is localized² near the origin; we would like to calculate the energy it radiates at time t_0 . Imagine a gigantic sphere, out at radius r (Fig. 11.1). The power passing through its surface is the integral of the Poynting vector:

$$P(r, t) = \oint \mathbf{S} \cdot d\mathbf{a} = \frac{1}{\mu_0} \oint (\mathbf{E} \times \mathbf{B}) \cdot d\mathbf{a}. \quad (11.1)$$

Because electromagnetic “news” travels at the speed of light,³ this energy actually left the source at the earlier time $t_0 = t - r/c$, so the power radiated is

$$P_{\text{rad}}(t_0) = \lim_{r \rightarrow \infty} P\left(r, t_0 + \frac{r}{c}\right) \quad (11.2)$$

(with t_0 held constant). This is energy (per unit time) that is carried away and never comes back.

Now, the area of the sphere is $4\pi r^2$, so for radiation to occur the Poynting vector must decrease (at large r) no faster than $1/r^2$ (if it went like $1/r^3$, for example, then P would go like $1/r$, and P_{rad} would be zero). According to Coulomb’s law, *electrostatic* fields fall off like $1/r^2$ (or even faster, if the total charge is zero), and the Biot–Savart law says that *magnetostatic* fields go like $1/r^2$ (or faster), which means that $S \sim 1/r^4$, for static configurations. So *static* sources do not radiate. But Jefimenko’s equations (Eqs. 10.36 and 10.38) indicate that *time-dependent* fields include terms (involving $\dot{\rho}$ and $\dot{\mathbf{J}}$) that go like $1/r$; these are the terms that are responsible for electromagnetic radiation.

¹ In this chapter, the word “radiation” is used in a restricted technical sense – it might better be called “radiation to infinity.” In everyday language the word has a broader connotation. We speak, for example, of radiation from a heat lamp or an X-ray machine. In this more general sense, electromagnetic “radiation” applies to any fields that transport energy – which is to say, fields whose Poynting vector is nonzero. There is nothing wrong with that language, but it is not how I am using the term here.

² For *nonlocalized* configurations, such as infinite planes, wires, or solenoids, the concept of “radiation” must be reformulated (Prob. 11.28).

³ More precisely, the fields depend on the status of the source at the retarded time.

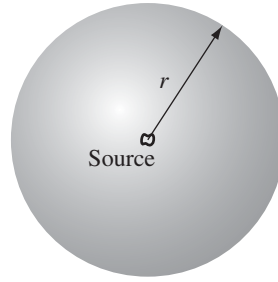


Fig. 11.1

The study of radiation, then, involves picking out the parts of $\mathbf{E}$ and $\mathbf{B}$ that go like $1/r$ at large distances from the source, constructing from them the $1/r^2$ term in $\mathbf{S}$, integrating over a large spherical⁴ surface, and taking the limit as $r \rightarrow \infty$. I'll carry through this procedure first for oscillating electric and magnetic dipoles; then, in Section 11.2, we'll consider the more difficult case of radiation from an accelerating point charge.

11.1.2 Electric Dipole Radiation

Picture two tiny metal spheres separated by a distance d and connected by a fine wire (Fig. 11.2); at time t the charge on the upper sphere is $q(t)$, and the charge on the lower sphere is $-q(t)$. Suppose that we drive the charge back and forth through the wire, from one end to the other, at an angular frequency ω :

$$q(t) = q_0 \cos(\omega t). \quad (11.3)$$

The result is an oscillating electric dipole:⁵

$$\mathbf{p}(t) = p_0 \cos(\omega t) \hat{\mathbf{z}}, \quad (11.4)$$

where

$$p_0 \equiv q_0 d$$

is the maximum value of the dipole moment.

The retarded potential (Eq. 10.26) is

$$V(\mathbf{r}, t) = \frac{1}{4\pi\epsilon_0} \left\{ \frac{q_0 \cos[\omega(t - r_+/c)]}{r_+} - \frac{q_0 \cos[\omega(t - r_-/c)]}{r_-} \right\}, \quad (11.5)$$

where, by the law of cosines,

$$r_{\pm} = \sqrt{r^2 \mp rd \cos \theta + (d/2)^2}. \quad (11.6)$$

⁴ It doesn't have to be a sphere, of course, but this makes the calculations a lot easier.

⁵ It might occur to you that a more natural model would consist of equal and opposite charges mounted on a spring, say, so that q is constant while d oscillates, instead of the other way around. Such a model would lead to the same result, but moving point charges are hard to work with, and the proposed formulation is much simpler.

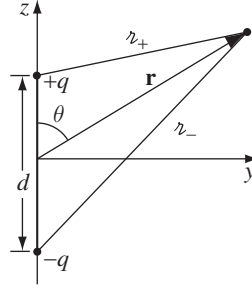


Fig. 11.2

Now, to make this *physical* dipole into a *perfect* dipole, we want the separation distance to be extremely small:

$$\text{approximation 1: } d \ll r. \quad (11.7)$$

Of course, if d is *zero* we get no potential at all; what we want is an expansion carried to *first order* in d . Thus

$$r_{\pm} \approx r \left(1 \mp \frac{d}{2r} \cos \theta \right). \quad (11.8)$$

It follows that

$$\frac{1}{r_{\pm}} \approx \frac{1}{r} \left(1 \pm \frac{d}{2r} \cos \theta \right), \quad (11.9)$$

and

$$\begin{aligned} & \cos[\omega(t - r_{\pm}/c)] \\ & \approx \cos\left[\omega(t - r/c) \pm \frac{\omega d}{2c} \cos \theta\right] \\ & = \cos[\omega(t - r/c)] \cos\left(\frac{\omega d}{2c} \cos \theta\right) \mp \sin[\omega(t - r/c)] \sin\left(\frac{\omega d}{2c} \cos \theta\right). \end{aligned}$$

In the perfect dipole limit we have, further,

$$\text{approximation 2: } d \ll \frac{c}{\omega}. \quad (11.10)$$

(Since waves of frequency ω have a wavelength $\lambda = 2\pi c/\omega$, this amounts to the requirement $d \ll \lambda$.) Under these conditions,

$$\cos[\omega(t - r_{\pm}/c)] \approx \cos[\omega(t - r/c)] \mp \frac{\omega d}{2c} \cos \theta \sin[\omega(t - r/c)]. \quad (11.11)$$

Putting Eqs. 11.9 and 11.11 into Eq. 11.5, we obtain the potential of an oscillating perfect dipole:

$$V(r, \theta, t) = \frac{p_0 \cos \theta}{4\pi\epsilon_0 r} \left\{ -\frac{\omega}{c} \sin[\omega(t - r/c)] + \frac{1}{r} \cos[\omega(t - r/c)] \right\}. \quad (11.12)$$

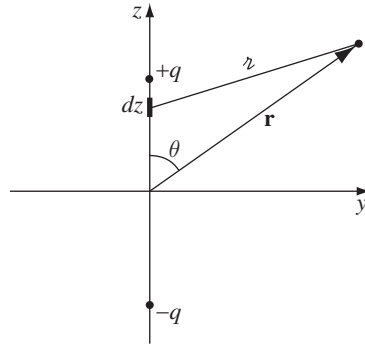


Fig. 11.3

In the static limit ($\omega \rightarrow 0$) the second term reproduces the old formula for the potential of a stationary dipole (Eq. 3.102):

$$V = \frac{p_0 \cos \theta}{4\pi\epsilon_0 r^2}.$$

This is not, however, the term that concerns us now; we are interested in the fields that survive at *large distances from the source*, in the so-called **radiation zone**:⁶

$$\text{approximation 3: } r \gg \frac{c}{\omega} \quad (11.13)$$

(or, in terms of the wavelength, $r \gg \lambda$). In this region the potential reduces to⁷

$$V(r, \theta, t) = -\frac{p_0 \omega}{4\pi\epsilon_0 c} \left(\frac{\cos \theta}{r} \right) \sin[\omega(t - r/c)]. \quad (11.14)$$

Meanwhile, the *vector* potential is determined by the current flowing in the wire:

$$\mathbf{I}(t) = \frac{dq}{dt} \hat{\mathbf{z}} = -q_0 \omega \sin(\omega t) \hat{\mathbf{z}}. \quad (11.15)$$

Referring to Fig. 11.3 (and Eq. 10.26),

$$\mathbf{A}(\mathbf{r}, t) = \frac{\mu_0}{4\pi} \int_{-d/2}^{d/2} \frac{-q_0 \omega \sin[\omega(t - r/c)] \hat{\mathbf{z}}}{r} dz. \quad (11.16)$$

Because the integration itself introduces a factor of d , we can, to first order, replace the integrand by its value at the center:

$$\mathbf{A}(r, \theta, t) = -\frac{\mu_0 p_0 \omega}{4\pi r} \sin[\omega(t - r/c)] \hat{\mathbf{z}}. \quad (11.17)$$

⁶ Note that approximations 2 and 3 subsume approximation 1; all together, we have $d \ll \lambda \ll r$.

⁷ I am dropping the fastidious $\approx$ in favor of $=$; it's not the *exact* potential, but it is the *relevant part* of that potential, if you're talking about radiation.

(Notice that whereas I implicitly used approximations 1 and 2, in keeping only the first order in d , Eq. 11.17 is not subject to approximation 3.)

From the potentials, it is a straightforward matter to compute the fields:

$$\begin{aligned}\nabla V &= \frac{\partial V}{\partial r} \hat{\mathbf{r}} + \frac{1}{r} \frac{\partial V}{\partial \theta} \hat{\boldsymbol{\theta}} \\ &= -\frac{p_0 \omega}{4\pi \epsilon_0 c} \left\{ \cos \theta \left(-\frac{1}{r^2} \sin[\omega(t-r/c)] - \frac{\omega}{rc} \cos[\omega(t-r/c)] \right) \hat{\mathbf{r}} \right. \\ &\quad \left. - \frac{\sin \theta}{r^2} \sin[\omega(t-r/c)] \hat{\boldsymbol{\theta}} \right\} \\ &\approx \frac{p_0 \omega^2}{4\pi \epsilon_0 c^2} \left(\frac{\cos \theta}{r} \right) \cos[\omega(t-r/c)] \hat{\mathbf{r}}.\end{aligned}$$

(I dropped the first and last terms, in accordance with approximation 3.) Likewise,

$$\frac{\partial \mathbf{A}}{\partial t} = -\frac{\mu_0 p_0 \omega^2}{4\pi r} \cos[\omega(t-r/c)] (\cos \theta \hat{\mathbf{r}} - \sin \theta \hat{\boldsymbol{\theta}}),$$

and therefore

$$\mathbf{E} = -\nabla V - \frac{\partial \mathbf{A}}{\partial t} = -\frac{\mu_0 p_0 \omega^2}{4\pi} \left(\frac{\sin \theta}{r} \right) \cos[\omega(t-r/c)] \hat{\boldsymbol{\theta}}. \quad (11.18)$$

Meanwhile,

$$\begin{aligned}\nabla \times \mathbf{A} &= \frac{1}{r} \left[\frac{\partial}{\partial r} (r A_\theta) - \frac{\partial A_r}{\partial \theta} \right] \hat{\boldsymbol{\phi}} \\ &= -\frac{\mu_0 p_0 \omega}{4\pi r} \left\{ \frac{\omega}{c} \sin \theta \cos[\omega(t-r/c)] + \frac{\sin \theta}{r} \sin[\omega(t-r/c)] \right\} \hat{\boldsymbol{\phi}}.\end{aligned}$$

The second term is again eliminated by approximation 3, so

$$\mathbf{B} = \nabla \times \mathbf{A} = -\frac{\mu_0 p_0 \omega^2}{4\pi c} \left(\frac{\sin \theta}{r} \right) \cos[\omega(t-r/c)] \hat{\boldsymbol{\phi}}. \quad (11.19)$$

Equations 11.18 and 11.19 represent monochromatic waves of frequency ω traveling in the radial direction at the speed of light. The fields are in phase, mutually perpendicular, and transverse; the ratio of their amplitudes is $E_0/B_0 = c$. All of this is precisely what we expect for electromagnetic waves in free space. (These are actually *spherical* waves, not plane waves, and their amplitude decreases like $1/r$ as they progress. But for large r , they are approximately plane over small regions – just as the surface of the Earth is reasonably flat, locally.)

The energy radiated by an oscillating electric dipole is determined by the Poynting vector:

$$\mathbf{S}(\mathbf{r}, t) = \frac{1}{\mu_0} (\mathbf{E} \times \mathbf{B}) = \frac{\mu_0}{c} \left\{ \frac{p_0 \omega^2}{4\pi} \left(\frac{\sin \theta}{r} \right) \cos[\omega(t-r/c)] \right\}^2 \hat{\mathbf{r}}. \quad (11.20)$$

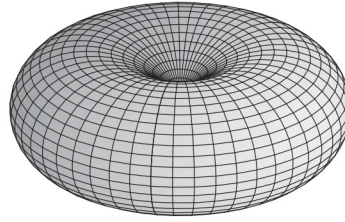


Fig. 11.4

The intensity is obtained by averaging (in time) over a complete cycle:

$$\langle \mathbf{S} \rangle = \left(\frac{\mu_0 p_0^2 \omega^4}{32\pi^2 c} \right) \frac{\sin^2 \theta}{r^2} \hat{\mathbf{r}}. \quad (11.21)$$

Notice that there is no radiation along the *axis* of the dipole (here $\sin \theta = 0$); the intensity profile⁸ takes the form of a donut, with its maximum in the equatorial plane (Fig. 11.4). The total power radiated is found by integrating $\langle \mathbf{S} \rangle$ over a sphere of radius r :

$$\langle P \rangle = \int \langle \mathbf{S} \rangle \cdot d\mathbf{a} = \frac{\mu_0 p_0^2 \omega^4}{32\pi^2 c} \int \frac{\sin^2 \theta}{r^2} r^2 \sin \theta d\theta d\phi = \frac{\mu_0 p_0^2 \omega^4}{12\pi c}. \quad (11.22)$$

Example 11.1. The strong frequency dependence of the power formula is what accounts for the blueness of the sky. Sunlight passing through the atmosphere stimulates atoms to oscillate as tiny dipoles. The incident solar radiation covers a broad range of frequencies (white light), but the energy absorbed and reradiated by the atmospheric dipoles is stronger at the higher frequencies because of the ω^4 in Eq. 11.22. It is more intense in the blue, then, than in the red. It is this reradiated light that you see when you look up in the sky – unless, of course, you’re staring directly at the Sun.

Because electromagnetic waves are transverse, the dipoles oscillate in a plane orthogonal to the Sun’s rays. In the celestial arc perpendicular to these rays, where the blueness is most pronounced, the dipoles oscillating along the line of sight send no radiation to the observer (because of the $\sin^2 \theta$ in Eq. 11.21); light received at this angle is therefore polarized perpendicular to the Sun’s rays (Fig. 11.5).

The redness of sunset is the other side of the same coin: Sunlight coming in along a tangent to the Earth’s surface must pass through a much longer stretch of atmosphere than sunlight coming from overhead (Fig. 11.6). Accordingly, much of the blue has been *removed* by scattering, and what’s left is red.

⁸ The radial coordinate in Fig. 11.4 represents the magnitude of $\langle \mathbf{S} \rangle$ in that direction.

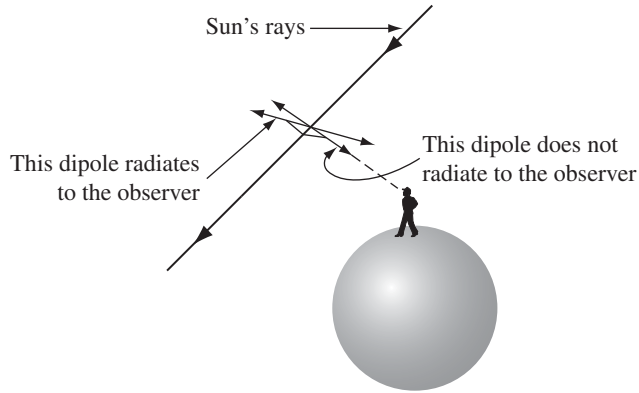


Fig. 11.5

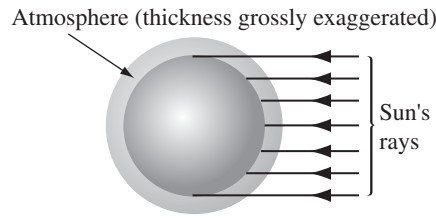


Fig. 11.6

Problem 11.1. Check that the retarded potentials of an oscillating dipole (Eqs. 11.12 and 11.17) satisfy the Lorenz gauge condition. Do *not* use approximation 3.

Problem 11.2. Equation 11.14 can be expressed in “coordinate-free” form by writing $p_0 \cos \theta = \mathbf{p}_0 \cdot \hat{\mathbf{r}}$. Do so, and likewise for Eqs. 11.17, 11.18, 11.19, and 11.21.

Problem 11.3. Find the **radiation resistance** of the wire joining the two ends of the dipole. (This is the resistance that would give the same average power loss – to heat – as the oscillating dipole in *fact* puts out in the form of radiation.) Show that $R = 790 (d/\lambda)^2 \Omega$, where λ is the wavelength of the radiation. For the wires in an ordinary radio (say, $d = 5$ cm), should you worry about the radiative contribution to the total resistance?

! **Problem 11.4.** A *rotating* electric dipole can be thought of as the superposition of two *oscillating* dipoles, one along the x -axis and the other along the y -axis (Fig. 11.7), with the latter out of phase by 90° :

$$\mathbf{p} = p_0 [\cos(\omega t) \hat{\mathbf{x}} + \sin(\omega t) \hat{\mathbf{y}}].$$

Using the principle of superposition and Eqs. 11.18 and 11.19 (perhaps in the form suggested by Prob. 11.2), find the fields of a rotating dipole. Also find the Poynting vector and the intensity of the radiation. Sketch the intensity profile as a function of the polar angle θ , and calculate the total power radiated. Does the answer seem reasonable? (Note that power, being *quadratic* in

the fields, does *not* satisfy the superposition principle. In this instance, however, it *seems* to. How do you account for this? You might want to refer to the “caution” at the end of Section 8.2.4.)

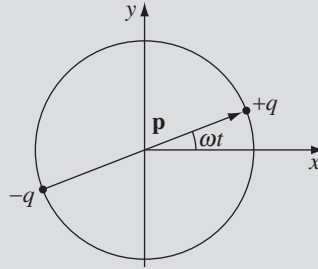


Fig. 11.7

11.1.3 Magnetic Dipole Radiation

Suppose now that we have a wire loop of radius b (Fig. 11.8), around which we drive an alternating current:

$$I(t) = I_0 \cos(\omega t). \quad (11.23)$$

This is a model for an oscillating *magnetic* dipole,

$$\mathbf{m}(t) = \pi b^2 I(t) \hat{\mathbf{z}} = m_0 \cos(\omega t) \hat{\mathbf{z}}, \quad (11.24)$$

where

$$m_0 \equiv \pi b^2 I_0 \quad (11.25)$$

is the maximum value of the magnetic dipole moment.

The loop is uncharged, so the scalar potential is zero. The retarded vector potential is

$$\mathbf{A}(\mathbf{r}, t) = \frac{\mu_0}{4\pi} \int \frac{I_0 \cos[\omega(t - r/c)]}{r} d\mathbf{l}'. \quad (11.26)$$

For a point $\mathbf{r}$ directly above the x -axis (Fig. 11.8), $\mathbf{A}$ must aim in the y direction, since the x components from symmetrically placed points on either side of the x -axis will cancel. Thus

$$\mathbf{A}(\mathbf{r}, t) = \frac{\mu_0 I_0 b}{4\pi} \hat{\mathbf{y}} \int_0^{2\pi} \frac{\cos[\omega(t - r/c)]}{r} \cos \phi' d\phi' \quad (11.27)$$

($\cos \phi'$ serves to pick out the y -component of $d\mathbf{l}'$). By the law of cosines,

$$r = \sqrt{r^2 + b^2 - 2rb \cos \psi},$$

where ψ is the angle between the vectors $\mathbf{r}$ and $\mathbf{b}$:

$$\mathbf{r} = r \sin \theta \hat{\mathbf{x}} + r \cos \theta \hat{\mathbf{z}}, \quad \mathbf{b} = b \cos \phi' \hat{\mathbf{x}} + b \sin \phi' \hat{\mathbf{y}}.$$

So $rb \cos \psi = \mathbf{r} \cdot \mathbf{b} = rb \sin \theta \cos \phi'$, and therefore

$$r = \sqrt{r^2 + b^2 - 2rb \sin \theta \cos \phi'}. \quad (11.28)$$

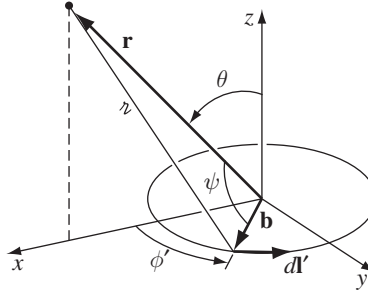


Fig. 11.8

For a “perfect” dipole, we want the loop to be extremely small:

$$\text{approximation 1: } b \ll r. \quad (11.29)$$

To first order in b , then,

$$r \approx r \left(1 - \frac{b}{r} \sin \theta \cos \phi' \right),$$

so

$$\frac{1}{r} \approx \frac{1}{r} \left(1 + \frac{b}{r} \sin \theta \cos \phi' \right) \quad (11.30)$$

and

$$\begin{aligned} \cos[\omega(t - r/c)] &\approx \cos\left[\omega(t - r/c) + \frac{\omega b}{c} \sin \theta \cos \phi'\right] \\ &= \cos[\omega(t - r/c)] \cos\left(\frac{\omega b}{c} \sin \theta \cos \phi'\right) \\ &\quad - \sin[\omega(t - r/c)] \sin\left(\frac{\omega b}{c} \sin \theta \cos \phi'\right). \end{aligned}$$

As before, we also assume the size of the dipole is small compared to the wavelength of the radiation:

$$\text{approximation 2: } b \ll \frac{c}{\omega}. \quad (11.31)$$

In that case,

$$\cos[\omega(t - r/c)] \approx \cos[\omega(t - r/c)] - \frac{\omega b}{c} \sin \theta \cos \phi' \sin[\omega(t - r/c)]. \quad (11.32)$$

Inserting Eqs. 11.30 and 11.32 into Eq. 11.27, and dropping the second-order term,

$$\begin{aligned} \mathbf{A}(\mathbf{r}, t) &\approx \frac{\mu_0 I_0 b}{4\pi r} \hat{\mathbf{y}} \int_0^{2\pi} \left\{ \cos[\omega(t - r/c)] + b \sin \theta \cos \phi' \right. \\ &\quad \times \left. \left(\frac{1}{r} \cos[\omega(t - r/c)] - \frac{\omega}{c} \sin[\omega(t - r/c)] \right) \right\} \cos \phi' d\phi'. \end{aligned}$$

The first term integrates to zero:

$$\int_0^{2\pi} \cos \phi' d\phi' = 0.$$

The second term involves the integral of cosine squared:

$$\int_0^{2\pi} \cos^2 \phi' d\phi' = \pi.$$

Putting this in, and noting that in general $\mathbf{A}$ points in the $\hat{\phi}$ direction, I conclude that the vector potential of an oscillating perfect magnetic dipole is

$$\mathbf{A}(r, \theta, t) = \frac{\mu_0 m_0}{4\pi} \left(\frac{\sin \theta}{r} \right) \left\{ \frac{1}{r} \cos[\omega(t - r/c)] - \frac{\omega}{c} \sin[\omega(t - r/c)] \right\} \hat{\phi}. \quad (11.33)$$

In the static limit ($\omega = 0$) we recover the familiar formula for the potential of a magnetic dipole (Eq. 5.87)

$$\mathbf{A}(r, \theta) = \frac{\mu_0 m_0 \sin \theta}{4\pi r^2} \hat{\phi}.$$

In the radiation zone,

$$\text{approximation 3: } r \gg \frac{c}{\omega}, \quad (11.34)$$

the first term in $\mathbf{A}$ is negligible, so

$$\mathbf{A}(r, \theta, t) = -\frac{\mu_0 m_0 \omega}{4\pi c} \left(\frac{\sin \theta}{r} \right) \sin[\omega(t - r/c)] \hat{\phi}. \quad (11.35)$$

From $\mathbf{A}$ we obtain the fields at large r :

$$\mathbf{E} = -\frac{\partial \mathbf{A}}{\partial t} = \frac{\mu_0 m_0 \omega^2}{4\pi c} \left(\frac{\sin \theta}{r} \right) \cos[\omega(t - r/c)] \hat{\phi}, \quad (11.36)$$

and

$$\mathbf{B} = \nabla \times \mathbf{A} = -\frac{\mu_0 m_0 \omega^2}{4\pi c^2} \left(\frac{\sin \theta}{r} \right) \cos[\omega(t - r/c)] \hat{\theta}. \quad (11.37)$$

(I used approximation 3 in calculating $\mathbf{B}$.) These fields are in phase, mutually perpendicular, and transverse to the direction of propagation ($\hat{\mathbf{r}}$), and the ratio of their amplitudes is $E_0/B_0 = c$, all of which is as expected for electromagnetic waves. They are, in fact, remarkably similar in structure to the fields of an oscillating *electric* dipole (Eqs. 11.18 and 11.19), only this time it is $\mathbf{B}$ that points in the $\hat{\theta}$ direction and $\mathbf{E}$ in the $\hat{\phi}$ direction, whereas for electric dipoles it's the other way around.

The energy flux for magnetic dipole radiation is

$$\mathbf{S}(\mathbf{r}, t) = \frac{1}{\mu_0} (\mathbf{E} \times \mathbf{B}) = \frac{\mu_0}{c} \left\{ \frac{m_0 \omega^2}{4\pi c} \left(\frac{\sin \theta}{r} \right) \cos[\omega(t - r/c)] \right\}^2 \hat{\mathbf{r}}, \quad (11.38)$$

the intensity is

$$\langle \mathbf{S} \rangle = \left(\frac{\mu_0 m_0^2 \omega^4}{32\pi^2 c^3} \right) \frac{\sin^2 \theta}{r^2} \hat{\mathbf{r}}, \quad (11.39)$$

and the total radiated power is

$$\langle P \rangle = \frac{\mu_0 m_0^2 \omega^4}{12\pi c^3}. \quad (11.40)$$

Once again, the intensity profile has the shape of a donut (Fig. 11.4), and the power radiated goes like ω^4 . There is, however, one important difference between electric and magnetic dipole radiation: for configurations with comparable dimensions, the power radiated electrically is enormously greater. Comparing Eqs. 11.22 and 11.40,

$$\frac{P_{\text{magnetic}}}{P_{\text{electric}}} = \left(\frac{m_0}{p_0 c} \right)^2, \quad (11.41)$$

where (remember) $m_0 = \pi b^2 I_0$, and $p_0 = q_0 d$. The amplitude of the current in the electrical case was $I_0 = q_0 \omega$ (Eq. 11.15). Setting $d = \pi b$, for the sake of comparison, I get

$$\frac{P_{\text{magnetic}}}{P_{\text{electric}}} = \left(\frac{\omega b}{c} \right)^2. \quad (11.42)$$

But $\omega b/c$ is precisely the quantity we assumed was very small (approximation 2), and here it appears *squared*. Ordinarily, then, one should expect electric dipole radiation to dominate. Only when the system is carefully contrived to exclude any electric contribution (as in the case just treated) will the magnetic dipole radiation reveal itself.

Problem 11.5. Calculate the electric and magnetic fields of an oscillating magnetic dipole *without* using approximation 3. [Do they look familiar? Compare Prob. 9.37.] Find the Poynting vector, and show that the intensity of the radiation is exactly the same as we got using approximation 3.

Problem 11.6. Find the radiation resistance (Prob. 11.3) for the oscillating magnetic dipole in Fig. 11.8. Express your answer in terms of λ and b , and compare the radiation resistance of the *electric* dipole. [Answer: $3 \times 10^5 (b/\lambda)^4 \Omega$.]

Problem 11.7. Use the “duality” transformation Prob. 7.68, together with the fields of an oscillating *electric* dipole (Eqs. 11.18 and 11.19), to determine the fields that would be produced by an oscillating “Gilbert” *magnetic* dipole (composed of equal and opposite magnetic charges, instead of an electric current loop). Compare Eqs. 11.36 and 11.37, and comment on the result.

11.1.4 Radiation from an Arbitrary Source

In the previous sections, we studied the radiation produced by two specific systems: oscillating electric dipoles and oscillating magnetic dipoles. Now I want to apply the same reasoning to a configuration of charge and current that is entirely arbitrary, except that it is localized within some finite volume near the origin (Fig. 11.9). The retarded scalar potential is

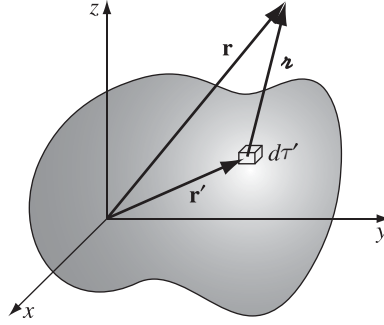


Fig. 11.9

$$V(\mathbf{r}, t) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\mathbf{r}', t - z/c)}{z} d\tau', \quad (11.43)$$

where

$$z = \sqrt{r^2 + r'^2 - 2\mathbf{r} \cdot \mathbf{r}'}. \quad (11.44)$$

As before, we shall assume that the field point $\mathbf{r}$ is far away, in comparison to the dimensions of the source:

$$\text{approximation 1: } r' \ll r. \quad (11.45)$$

(Actually, r' is a variable of integration; approximation 1 means that the *maximum* value of r' , as it ranges over the source, is much less than r .) On this assumption,

$$z \approx r \left(1 - \frac{\mathbf{r} \cdot \mathbf{r}'}{r^2} \right), \quad (11.46)$$

so

$$\frac{1}{z} \approx \frac{1}{r} \left(1 + \frac{\mathbf{r} \cdot \mathbf{r}'}{r^2} \right) \quad (11.47)$$

and

$$\rho(\mathbf{r}', t - z/c) \approx \rho\left(\mathbf{r}', t - \frac{r}{c} + \frac{\hat{\mathbf{r}} \cdot \mathbf{r}'}{c}\right).$$

Expanding ρ as a Taylor series in t about the retarded time at the origin,

$$t_0 \equiv t - \frac{r}{c}, \quad (11.48)$$

we have

$$\rho(\mathbf{r}', t - z/c) \approx \rho(\mathbf{r}', t_0) + \dot{\rho}(\mathbf{r}', t_0) \left(\frac{\hat{\mathbf{r}} \cdot \mathbf{r}'}{c} \right) + \dots, \quad (11.49)$$

where the dot signifies differentiation with respect to time. The next terms in the series would be

$$\frac{1}{2} \ddot{\rho} \left(\frac{\hat{\mathbf{r}} \cdot \mathbf{r}'}{c} \right)^2, \quad \frac{1}{3!} \dddot{\rho} \left(\frac{\hat{\mathbf{r}} \cdot \mathbf{r}'}{c} \right)^3, \dots$$

We can afford to drop them, provided

$$\text{approximation 2: } r' \ll \frac{c}{|\ddot{\rho}/\dot{\rho}|}, \quad \frac{c}{|\ddot{\rho}/\dot{\rho}|^{1/2}}, \quad \frac{c}{|\ddot{\rho}/\dot{\rho}|^{1/3}}, \dots \quad (11.50)$$

For an oscillating system, each of these ratios is $cl\omega$, and we recover the old approximation 2. In the general case it's more difficult to interpret Eq. 11.50, but as a *procedural* matter approximations 1 and 2 amount to *keeping only the first-order terms in r'* .

Putting Eqs. 11.47 and 11.49 into the formula for V (Eq. 11.43), and again discarding the second-order term,

$$V(\mathbf{r}, t) \approx \frac{1}{4\pi\epsilon_0 r} \left[\int \rho(\mathbf{r}', t_0) d\tau' + \frac{\hat{\mathbf{r}}}{r} \cdot \int \mathbf{r}' \rho(\mathbf{r}', t_0) d\tau' + \frac{\hat{\mathbf{r}}}{c} \cdot \frac{d}{dt} \int \mathbf{r}' \rho(\mathbf{r}', t_0) d\tau' \right].$$

The first integral is simply the total charge, Q , at time t_0 . Because charge is conserved, Q is independent of time. The other two integrals represent the electric dipole moment at time t_0 . Thus

$$V(\mathbf{r}, t) \approx \frac{1}{4\pi\epsilon_0} \left[\frac{Q}{r} + \frac{\hat{\mathbf{r}} \cdot \mathbf{p}(t_0)}{r^2} + \frac{\hat{\mathbf{r}} \cdot \dot{\mathbf{p}}(t_0)}{rc} \right]. \quad (11.51)$$

In the static case, the first two terms are the monopole and dipole contributions to the multipole expansion for V ; the third term, of course, would not be present.

Meanwhile, the vector potential is

$$\mathbf{A}(\mathbf{r}, t) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{r}', t - r/c)}{r} d\tau'. \quad (11.52)$$

As you'll see in a moment, to first order in r' it suffices to replace r by r in the integrand:

$$\mathbf{A}(\mathbf{r}, t) \approx \frac{\mu_0}{4\pi r} \int \mathbf{J}(\mathbf{r}', t_0) d\tau'. \quad (11.53)$$

According to Prob. 5.7, the integral of $\mathbf{J}$ is the time derivative of the dipole moment, so

$$\mathbf{A}(\mathbf{r}, t) \approx \frac{\mu_0}{4\pi} \frac{\dot{\mathbf{p}}(t_0)}{r}. \quad (11.54)$$

Now you see why it was unnecessary to carry the approximation of r beyond the zeroth order ($r \approx r$): $\mathbf{p}$ is *already* first order in r' , and any refinements would be corrections of *second* order (or higher).

Next we must calculate the fields. Once again, we are interested in the radiation zone (that is, in the fields that survive at large distances from the source), so we keep only those terms that go like $1/r$:

$$\textbf{approximation 3:} \quad \text{discard } 1/r^2 \text{ terms in } \mathbf{E} \text{ and } \mathbf{B}. \quad (11.55)$$

For instance, the Coulomb field,

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} \hat{\mathbf{r}},$$

coming from the first term in Eq. 11.51, does not contribute to the electromagnetic radiation. In fact, the radiation comes entirely from those terms in which we differentiate the argument t_0 . From Eq. 11.48 it follows that

$$\nabla t_0 = -\frac{1}{c} \nabla r = -\frac{1}{c} \hat{\mathbf{r}},$$

and hence

$$\nabla V \approx \nabla \left[\frac{1}{4\pi\epsilon_0} \frac{\hat{\mathbf{r}} \cdot \dot{\mathbf{p}}(t_0)}{rc} \right] \approx \frac{1}{4\pi\epsilon_0} \left[\frac{\hat{\mathbf{r}} \cdot \ddot{\mathbf{p}}(t_0)}{rc} \right] \nabla t_0 = -\frac{1}{4\pi\epsilon_0 c^2} \frac{[\hat{\mathbf{r}} \cdot \ddot{\mathbf{p}}(t_0)]}{r} \hat{\mathbf{r}}.$$

Similarly,

$$\nabla \times \mathbf{A} \approx \frac{\mu_0}{4\pi r} [\nabla \times \dot{\mathbf{p}}(t_0)] = \frac{\mu_0}{4\pi r} [(\nabla t_0) \times \ddot{\mathbf{p}}(t_0)] = -\frac{\mu_0}{4\pi rc} [\hat{\mathbf{r}} \times \ddot{\mathbf{p}}(t_0)],$$

while

$$\frac{\partial \mathbf{A}}{\partial t} \approx \frac{\mu_0}{4\pi} \frac{\ddot{\mathbf{p}}(t_0)}{r}.$$

So

$$\boxed{\mathbf{E}(\mathbf{r}, t) \approx \frac{\mu_0}{4\pi r} [(\hat{\mathbf{r}} \cdot \ddot{\mathbf{p}})\hat{\mathbf{r}} - \ddot{\mathbf{p}}] = \frac{\mu_0}{4\pi r} [\hat{\mathbf{r}} \times (\hat{\mathbf{r}} \times \ddot{\mathbf{p}})],} \quad (11.56)$$

where $\ddot{\mathbf{p}}$ is evaluated at time $t_0 = t - r/c$, and

$$\boxed{\mathbf{B}(\mathbf{r}, t) \approx -\frac{\mu_0}{4\pi rc} [\hat{\mathbf{r}} \times \ddot{\mathbf{p}}].} \quad (11.57)$$

In particular, if we use spherical polar coordinates, with the z -axis in the direction of $\ddot{\mathbf{p}}(t_0)$, then

$$\left. \begin{aligned} \mathbf{E}(r, \theta, t) &\approx \frac{\mu_0 \ddot{p}(t_0)}{4\pi} \left(\frac{\sin \theta}{r} \right) \hat{\boldsymbol{\theta}}, \\ \mathbf{B}(r, \theta, t) &\approx \frac{\mu_0 \ddot{p}(t_0)}{4\pi c} \left(\frac{\sin \theta}{r} \right) \hat{\boldsymbol{\phi}}. \end{aligned} \right\} \quad (11.58)$$

The Poynting vector is

$$\mathbf{S}(\mathbf{r}, t) \approx \frac{1}{\mu_0} (\mathbf{E} \times \mathbf{B}) = \frac{\mu_0}{16\pi^2 c} [\ddot{p}(t_0)]^2 \left(\frac{\sin^2 \theta}{r^2} \right) \hat{\mathbf{r}}, \quad (11.59)$$

the power passing through a giant spherical surface at radius r is

$$P(r, t) = \oint \mathbf{S}(\mathbf{r}, t) \cdot d\mathbf{a} = \frac{\mu_0}{6\pi c} \left[\ddot{p} \left(t - \frac{r}{c} \right) \right]^2,$$

and the total radiated power (Eq. 11.2) is

$$P_{\text{rad}}(t_0) \approx \frac{\mu_0}{6\pi c} [\ddot{p}(t_0)]^2. \quad (11.60)$$

Notice that $\mathbf{E}$ and $\mathbf{B}$ are mutually perpendicular, transverse to the direction of propagation ($\hat{\mathbf{r}}$), and in the ratio $E/B = c$, as always for radiation fields.

Example 11.2.

(a) In the case of an oscillating electric dipole,

$$p(t) = p_0 \cos(\omega t), \quad \ddot{p}(t) = -\omega^2 p_0 \cos(\omega t),$$

and we recover the results of Section 11.1.2.

(b) For a single point charge q , the dipole moment is

$$\mathbf{p}(t) = q\mathbf{d}(t),$$

where $\mathbf{d}$ is the position of q with respect to the origin. Accordingly,

$$\ddot{\mathbf{p}}(t) = q\mathbf{a}(t),$$

where $\mathbf{a}$ is the acceleration of the charge. In this case the power radiated (Eq. 11.60) is

$$P = \frac{\mu_0 q^2 a^2}{6\pi c}. \quad (11.61)$$

This is the famous **Larmor formula**; I'll derive it again, by rather different means, in the next section. Notice that the power radiated by a point charge is proportional to the *square* of its *acceleration*.

What I have done in this section amounts to a multipole expansion of the retarded potentials, carried to the lowest order in r' that is capable of producing electromagnetic radiation (fields that go like $1/r$). This turns out to be the electric dipole term. Because charge is conserved, an electric *monopole* does not radiate – if charge were *not* conserved, the first term in Eq. 11.51 would read

$$V_{\text{mono}} = \frac{1}{4\pi\epsilon_0} \frac{Q(t_0)}{r},$$

and we would get a monopole field proportional to $1/r$:

$$\mathbf{E}_{\text{mono}} = \frac{1}{4\pi\epsilon_0 c} \frac{\dot{Q}(t_0)}{r} \hat{\mathbf{r}}.$$

You might think that a charged sphere whose radius oscillates in and out would radiate, but it *doesn't* – the field outside, according to Gauss's law, is exactly $(Q/4\pi\epsilon_0 r^2)\hat{\mathbf{r}}$, regardless of the fluctuations in size. (In the acoustical analog, by the way, monopoles *do* radiate: witness the croak of a bullfrog.)

If the electric dipole moment should happen to vanish (or, at any rate, if its second time derivative is zero), then there is no electric dipole radiation, and one must look to the next term: the one of *second* order in r' . As it happens, this term can be separated into two parts, one of which is related to the *magnetic* dipole moment of the source, the other to its electric *quadrupole* moment. (The former is a generalization of the magnetic dipole radiation we considered in Section 11.1.3.) If the magnetic dipole and electric quadrupole contributions vanish, the $(r')^3$ term must be considered. This yields magnetic quadrupole and electric octopole radiation ... and so it goes.

Problem 11.8. A parallel-plate capacitor C , with plate separation d , is given an initial charge $(\pm)Q_0$. It is then connected to a resistor R , and discharges, $Q(t) = Q_0 e^{-t/RC}$.

- (a) What fraction of its initial energy ($Q_0^2/2C$) does it radiate away?
 (b) If $C = 1 \text{ pF}$, $R = 1000 \Omega$, and $d = 0.1 \text{ mm}$, what is the actual *number*? In electronics we don't ordinarily worry about radiative losses; does that seem reasonable, in this case?

Problem 11.9. Apply Eqs. 11.59 and 11.60 to the rotating dipole of Prob. 11.4. Explain any apparent discrepancies with your previous answer.

Problem 11.10. An insulating circular ring (radius b) lies in the xy plane, centered at the origin. It carries a linear charge density $\lambda = \lambda_0 \sin \phi$, where λ_0 is constant and ϕ is the usual azimuthal angle. The ring is now set spinning at a constant angular velocity ω about the z -axis. Calculate the power radiated.

- ! **Problem 11.11.** A current $I(t)$ flows around the circular ring in Fig. 11.8. Derive the general formula for the power radiated (analogous to Eq. 11.60), expressing your answer in terms of the magnetic dipole moment, $m(t)$, of the loop. [Answer: $P = \mu_0 \ddot{m}^2 / 6\pi c^3$.]

11.2 Power Radiated by a Point Charge

11.2.1 The Larmor Formula

In Chapter 10 we derived the fields of a point charge q in arbitrary motion (Eqs. 10.72 and 10.73):

$$\mathbf{E}(\mathbf{r}, t) = \frac{q}{4\pi\epsilon_0} \frac{\hat{\mathbf{z}}}{(\hat{\mathbf{z}} \cdot \mathbf{u})^3} [(c^2 - v^2)\mathbf{u} + \hat{\mathbf{z}} \times (\mathbf{u} \times \mathbf{a})], \quad (11.62)$$

where $\mathbf{u} = c\hat{\mathbf{z}} - \mathbf{v}$, and

$$\mathbf{B}(\mathbf{r}, t) = \frac{1}{c} \hat{\mathbf{z}} \times \mathbf{E}(\mathbf{r}, t). \quad (11.63)$$

The first term in Eq. 11.62 is the **velocity field**, and the second one (with the triple cross product) is the **acceleration field**.

The Poynting vector is

$$\mathbf{S} = \frac{1}{\mu_0} (\mathbf{E} \times \mathbf{B}) = \frac{1}{\mu_0 c} [\mathbf{E} \times (\hat{\mathbf{z}} \times \mathbf{E})] = \frac{1}{\mu_0 c} [E^2 \hat{\mathbf{z}} - (\hat{\mathbf{z}} \cdot \mathbf{E})\mathbf{E}]. \quad (11.64)$$

However, not all of this energy flux constitutes *radiation*; some of it is just field energy carried along by the particle as it moves. The *radiated* energy is the stuff that, in effect, *detaches* itself from the charge and propagates off to infinity. (It's like flies breeding on a garbage truck: some of them hover around the truck as it makes its rounds; others fly away and never come back.) To calculate the total power *radiated* by the particle at time t , we draw a huge sphere of radius $\mathcal{r}$ (Fig. 11.10), centered at the position of the particle (at time t). At the later time

$$t_s \equiv t + \frac{\mathcal{r}}{c}, \quad (11.65)$$

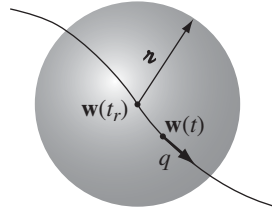


Fig. 11.10

the radiation reaches the sphere, and at that moment we integrate the Poynting vector over the surface.⁹ Notice that for all points on the sphere (at time t_s), $t = t_s - r/c$ is precisely the retarded time (Eq. 10.25), so in evaluating the fields we'll need the position, velocity, and acceleration of the charge *at time t* , when the radiation left the source.

Now, the area of the sphere is proportional to r^2 , so any term in $\mathbf{S}$ that goes like $1/r^2$ will yield a finite answer, but terms like $1/r^3$ or $1/r^4$ will contribute nothing in the limit $r \rightarrow \infty$. For this reason, only the *acceleration* fields represent true radiation (hence their other name, **radiation fields**):

$$\mathbf{E}_{\text{rad}} = \frac{q}{4\pi\epsilon_0} \frac{r}{(r \cdot \mathbf{u})^3} [\mathbf{r} \times (\mathbf{u} \times \mathbf{a})]. \quad (11.66)$$

The velocity fields carry energy, to be sure, and as the charge moves this energy is dragged along – but it's not *radiation*. (It's like the flies that stay with the garbage truck.) Now $\mathbf{E}_{\text{rad}}$ is perpendicular to $\hat{\mathbf{r}}$, so the second term in Eq. 11.64 vanishes:

$$\mathbf{S}_{\text{rad}} = \frac{1}{\mu_0 c} E_{\text{rad}}^2 \hat{\mathbf{r}}. \quad (11.67)$$

If the charge is instantaneously *at rest* (at time t), then $\mathbf{u} = c\hat{\mathbf{r}}$, and

$$\mathbf{E}_{\text{rad}} = \frac{q}{4\pi\epsilon_0 c^2 r} [\hat{\mathbf{r}} \times (\hat{\mathbf{r}} \times \mathbf{a})] = \frac{\mu_0 q}{4\pi r} [(\hat{\mathbf{r}} \cdot \mathbf{a})\hat{\mathbf{r}} - \mathbf{a}]. \quad (11.68)$$

In that case,

$$\mathbf{S}_{\text{rad}} = \frac{1}{\mu_0 c} \left(\frac{\mu_0 q}{4\pi r} \right)^2 [a^2 - (\hat{\mathbf{r}} \cdot \mathbf{a})^2] \hat{\mathbf{r}} = \frac{\mu_0 q^2 a^2}{16\pi^2 c} \left(\frac{\sin^2 \theta}{r^2} \right) \hat{\mathbf{r}}, \quad (11.69)$$

where θ is the angle between $\hat{\mathbf{r}}$ and $\mathbf{a}$. No power is radiated in the forward or backward direction – rather, it is emitted in a donut about the direction of instantaneous acceleration (Fig. 11.11). The *total* power radiated (at time t) is

$$P = \oint \mathbf{S}_{\text{rad}} \cdot d\mathbf{a} = \frac{\mu_0 q^2 a^2}{16\pi^2 c} \int \frac{\sin^2 \theta}{r^2} r^2 \sin \theta d\theta d\phi$$

⁹ There is a subtle change in strategy here: in Section 11.1 we worked from a fixed point (the origin), but here it is more appropriate to use the (moving) location of the charge. The implications of this change in perspective will become clearer in a moment.

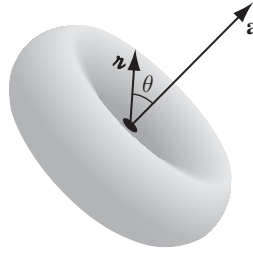


Fig. 11.11

(where a is the acceleration at time t) or

$$P = \frac{\mu_0 q^2 a^2}{6\pi c}. \quad (11.70)$$

This, again, is the **Larmor formula**, which we obtained earlier by a quite different route (Eq. 11.61). Although I *derived* Eqs. 11.69 and 11.70 on the assumption that $v = 0$, they actually hold to good approximation as long as $v \ll c$.¹⁰

I'll treat the general case ($v \neq 0$) in a moment, but first I want to call your attention to the connection between radiation and *acceleration*, as spotlighted by the Larmor formula. A particle at rest does not radiate; nor does a particle moving at constant velocity ... but if it's *accelerating*, then it radiates. *Wait a minute!* What about a uniformly charged circular ring (radius R) rotating at a constant angular velocity (ω) about its axis? That's a *steady* current, the electric and magnetic fields are *static*, and it certainly doesn't radiate. And yet, every point on the ring is accelerating ($a = \omega^2 R$). Or consider a uniformly charged spherical shell (radius R) that "breathes" in and out sinusoidally ($R(t) = R_0 \cos \omega t$); for $r > R_0$ the electric field is Coulombic, and the magnetic field is zero everywhere. No radiation ... and yet every point on the shell is accelerating. Doesn't the Larmor formula say that acceleration generates radiation? *Yes*: but the Larmor formula applies to *point* charges, whereas these are *extended* objects. Remember, the Poynting vector does not satisfy the superposition principle – the total power radiated is *not* the sum of the powers that would be radiated by each electron separately. If you like, what we have in these examples is perfect destructive interference in every direction.¹¹

11.2.2 The Liénard Formula

The exact treatment of the general case ($v \neq 0$) is harder, both for the obvious reason that $\mathbf{E}_{\text{rad}}$ is more complicated, and also for the subtler reason

¹⁰ In the context of special relativity, the condition $v = 0$ simply represents an astute choice of reference frame, with no essential loss of generality. If you can figure out how P transforms, you can *deduce* the general (Liénard) result from the $v = 0$ (Larmor) formula (see Prob. 12.74).

¹¹ T. A. Abbott and D. J. Griffiths, *Am. J. Phys.* **53**, 1203 (1985).

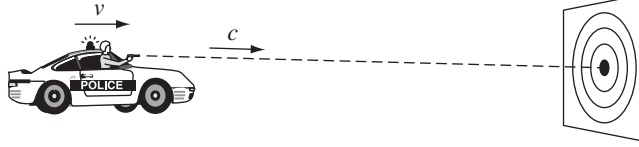


Fig. 11.12

that S_{rad} , the rate at which energy passes through the sphere (Fig. 11.10) is *not* the same as the rate at which that energy left the particle. To understand why, imagine someone is firing a stream of bullets out the window of a moving car (Fig. 11.12). The rate N_t at which the bullets strike a stationary target is not the same as the rate N_g at which they left the gun, because of the motion of the car. In fact, you can easily check that $N_g = (1 - v/c)N_t$, if the car is moving toward the target, and

$$N_g = \left(1 - \frac{\hat{\mathbf{z}} \cdot \mathbf{v}}{c}\right) N_t$$

for arbitrary directions (here $\mathbf{v}$ is the velocity of the car, c is the speed of the bullets – relative to the ground – and $\hat{\mathbf{z}}$ is a unit vector from car to target). More formally, if dW/dt_s is the rate at which energy passes through the sphere at radius z , then the rate at which energy left the charge was

$$\frac{dW}{dt} = \frac{dW}{dt_s} \frac{\partial t_s}{\partial t} = \left(\frac{\mathbf{z} \cdot \mathbf{u}}{zc}\right) \frac{dW}{dt_s}. \quad (11.71)$$

(I used Eq. 10.78 to get $\partial t/\partial t_s$.) But

$$\frac{\mathbf{z} \cdot \mathbf{u}}{zc} = 1 - \frac{\hat{\mathbf{z}} \cdot \mathbf{v}}{c},$$

which is precisely the ratio of N_g to N_t ; it's a purely geometrical factor (the same as in the Doppler effect).

The power radiated by the particle into a patch of area $z^2 \sin \theta d\theta d\phi = z^2 d\Omega$ on the sphere is therefore given by

$$\frac{dP}{d\Omega} = \left(\frac{\mathbf{z} \cdot \mathbf{u}}{zc}\right) \frac{1}{\mu_0 c} E_{\text{rad}}^2 z^2 = \frac{q^2}{16\pi^2 \epsilon_0} \frac{|\hat{\mathbf{z}} \times (\mathbf{u} \times \mathbf{a})|^2}{(\hat{\mathbf{z}} \cdot \mathbf{u})^5}, \quad (11.72)$$

where $d\Omega = \sin \theta d\theta d\phi$ is the **solid angle** into which this power is radiated. Integrating over θ and ϕ to get the total power radiated is no picnic, and for now I shall simply quote the answer:

$$P = \int \frac{dP}{d\Omega} \sin \theta d\theta d\phi = \frac{\mu_0 q^2 \gamma^6}{6\pi c} \left(a^2 - \left|\frac{\mathbf{v} \times \mathbf{a}}{c}\right|^2\right), \quad (11.73)$$

where $\gamma \equiv 1/\sqrt{1 - v^2/c^2}$. This is **Liénard's generalization** of the Larmor formula (to which it reduces when $v \ll c$). The factor γ^6 means that the radiated power increases enormously as the particle velocity approaches the speed of light.

Example 11.3. Suppose that $\mathbf{v}$ and $\mathbf{a}$ are instantaneously collinear (at time t), as (for example) in straight-line motion. Find the angular distribution of the radiation (Eq. 11.72) and the total power emitted.

Solution. In this case $(\mathbf{u} \times \mathbf{a}) = c(\hat{\mathbf{z}} \times \mathbf{a})$, so

$$\frac{dP}{d\Omega} = \frac{q^2 c^2}{16\pi^2 \epsilon_0} \frac{|\hat{\mathbf{z}} \times (\hat{\mathbf{z}} \times \mathbf{a})|^2}{(c - \hat{\mathbf{z}} \cdot \mathbf{v})^5}.$$

Now

$$\hat{\mathbf{z}} \times (\hat{\mathbf{z}} \times \mathbf{a}) = (\hat{\mathbf{z}} \cdot \mathbf{a})\hat{\mathbf{z}} - \mathbf{a}, \quad \text{so } |\hat{\mathbf{z}} \times (\hat{\mathbf{z}} \times \mathbf{a})|^2 = a^2 - (\hat{\mathbf{z}} \cdot \mathbf{a})^2.$$

In particular, if we let the z -axis point along $\mathbf{v}$, then

$$\frac{dP}{d\Omega} = \frac{\mu_0 q^2 a^2}{16\pi^2 c} \frac{\sin^2 \theta}{(1 - \beta \cos \theta)^5}, \quad (11.74)$$

where $\beta \equiv v/c$. This is consistent, of course, with Eq. 11.69, in the case $v = 0$. However, for very large v ($\beta \approx 1$) the donut of radiation (Fig. 11.11) is stretched out and pushed forward by the factor $(1 - \beta \cos \theta)^{-5}$, as indicated in Fig. 11.13. Although there is still no radiation in *precisely* the forward direction, most of it is concentrated within an increasingly narrow cone *about* the forward direction (see Prob. 11.15).

The *total* power emitted is found by integrating Eq. 11.74 over all angles:

$$P = \frac{\mu_0 q^2 a^2}{16\pi^2 c} \int \frac{\sin^2 \theta}{(1 - \beta \cos \theta)^5} \sin \theta \, d\theta \, d\phi.$$

The ϕ integral is 2π ; the θ integral is simplified by the substitution $x \equiv \cos \theta$:

$$P = \frac{\mu_0 q^2 a^2}{8\pi c} \int_{-1}^{+1} \frac{(1 - x^2)}{(1 - \beta x)^5} dx.$$

Integration by parts yields $\frac{4}{3}(1 - \beta^2)^{-3}$, and I conclude that

$$P = \frac{\mu_0 q^2 a^2 \gamma^6}{6\pi c}. \quad (11.75)$$

This result is consistent with the Liénard formula (Eq. 11.73), for the case of collinear $\mathbf{v}$ and $\mathbf{a}$. Notice that the angular distribution of the radiation is the same whether the particle is *accelerating* or *decelerating*; it only depends on the *square* of a , and is concentrated in the forward direction (with respect to the velocity) in either case. When a high speed electron hits a metal target it rapidly decelerates, giving off what is called **bremstrahlung**, or “braking radiation.” What I have described in this example is essentially the classical theory of bremstrahlung.

Problem 11.12. An electron is released from rest and falls under the influence of gravity. In the first centimeter, what fraction of the potential energy lost is radiated away?

Problem 11.13. A positive charge q is fired head-on at a distant positive charge Q (which is held stationary), with an initial velocity v_0 . It comes in, decelerates to $v = 0$, and returns out to infinity. What fraction of its initial energy ($\frac{1}{2}mv_0^2$) is radiated away? Assume $v_0 \ll c$, and that you can safely ignore the effect of radiative losses on the motion of the particle. [Answer: $(16/45)(q/Q)(v_0/c)^3$.]

Problem 11.14. In Bohr's theory of hydrogen, the electron in its ground state was supposed to travel in a circle of radius 5×10^{-11} m, held in orbit by the Coulomb attraction of the proton. According to classical electrodynamics, this electron should radiate, and hence spiral in to the nucleus. Show that $v \ll c$ for most of the trip (so you can use the Larmor formula), and calculate the lifespan of Bohr's atom. (Assume each revolution is essentially circular.)

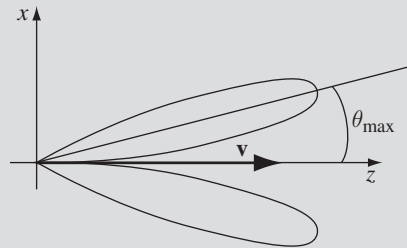


Fig. 11.13

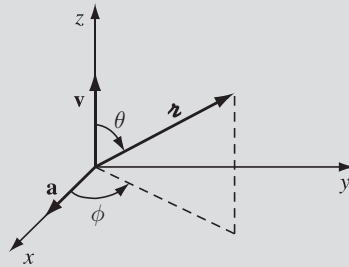


Fig. 11.14

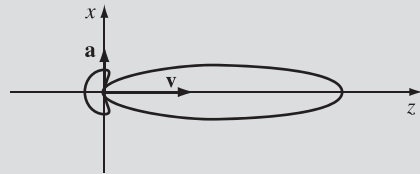


Fig. 11.15

Problem 11.15. Find the angle $\theta_{\max}$ at which the maximum radiation is emitted, in Ex. 11.3 (Fig. 11.13). Show that for ultrarelativistic speeds (v close to c),

$$\theta_{\max} \approx \sqrt{(1 - \beta)/2}.$$

What is the intensity of the radiation in this maximal direction (in the ultrarelativistic case), in proportion to the same quantity for a particle instantaneously at rest? Give your answer in terms of γ .

! **Problem 11.16.** In Ex. 11.3 we assumed the velocity and acceleration were (instantaneously, at least) *collinear*. Carry out the same analysis for the case where they are *perpendicular*. Choose your axes so that $\mathbf{v}$ lies along the z -axis and $\mathbf{a}$ along the x -axis (Fig. 11.14), so that $\mathbf{v} = v\hat{z}$,

$\mathbf{a} = a\hat{\mathbf{x}}$, and $\hat{\mathbf{z}} = \sin \theta \cos \phi \hat{\mathbf{x}} + \sin \theta \sin \phi \hat{\mathbf{y}} + \cos \theta \hat{\mathbf{z}}$. Check that P is consistent with the Liénard formula. [Answer:

$$\frac{dP}{d\Omega} = \frac{\mu_0 q^2 a^2}{16\pi^2 c} \frac{[(1 - \beta \cos \theta)^2 - (1 - \beta^2) \sin^2 \theta \cos^2 \phi]}{(1 - \beta \cos \theta)^5}, \quad P = \frac{\mu_0 q^2 a^2 \gamma^4}{6\pi c}.$$

For relativistic velocities ($\beta \approx 1$) the radiation is again sharply peaked in the forward direction (Fig. 11.15). The most important application of these formulas is to *circular* motion – in this case the radiation is called **synchrotron radiation**. For a relativistic electron, the radiation sweeps around like a locomotive’s headlight as the particle moves.]

11.3 The Radiation Reaction

When a charged particle accelerates, it radiates. This radiation carries off energy, which must come at the expense of some other form – kinetic, potential, or energy stored in the nonradiative fields of the particle. Under the influence of a given force, therefore, the acceleration of a charged particle is typically less than – or at any rate *different* from – that of a neutral particle of the same mass. Evidently, the fields exert a force ($\mathbf{F}_{\text{rad}}$) back on the charge – a *recoil* force, rather like that of a bullet on a gun. In the next section I’ll derive this **radiation reaction** force from conservation of energy. Then, in the following section, I’ll show you the actual *mechanism* responsible, and derive the reaction force again in the context of a simple model.

11.3.1 The Abraham–Lorentz Formula

For a nonrelativistic particle ($v \ll c$), the total power radiated is given by the Larmor formula (Eq. 11.70):

$$P = \frac{\mu_0 q^2 a^2}{6\pi c}. \quad (11.76)$$

Conservation of energy suggests that this is also the rate at which the particle *loses* energy, under the influence of the radiation reaction force $\mathbf{F}_{\text{rad}}$:

$$\mathbf{F}_{\text{rad}} \cdot \mathbf{v} = -\frac{\mu_0 q^2 a^2}{6\pi c}. \quad (11.77)$$

I say “suggests” advisedly, because this equation is actually *wrong*. For we calculated the radiated power by integrating the Poynting vector over a sphere of “infinite” radius; in this calculation the *velocity* fields played no part, since they fall off too rapidly as a function of r to make any contribution. But these “local” fields *do* carry energy – they just don’t transport it out to infinity. As the particle accelerates and decelerates, energy is exchanged between it and the nearby fields, at the same time as energy is irretrievably radiated away by the acceleration fields. Equation 11.77

accounts only for the latter, but if we want to know the recoil force exerted by the fields on the charge, we need to consider the *total* power lost at any instant, not just the portion that eventually escapes in the form of radiation. (The term “radiation reaction” is a misnomer. We should really call it the *field reaction*. In fact, we’ll soon see that $\mathbf{F}_{\text{rad}}$ is determined by the *time derivative* of the acceleration and can be nonzero even when the acceleration itself is instantaneously zero, and hence the particle is not radiating.)

The energy lost by the particle in any given time interval, then, must equal the energy carried away by the radiation *plus* whatever extra energy has been pumped into the nearby fields.¹² However, if we agree to consider only intervals over which the system returns to its initial state, then the energy in the velocity fields is the same at both ends, and the only *net* loss is in the form of radiation. Thus Eq. 11.77, while incorrect *instantaneously*, is valid on the *average*:

$$\int_{t_1}^{t_2} \mathbf{F}_{\text{rad}} \cdot \mathbf{v} dt = -\frac{\mu_0 q^2}{6\pi c} \int_{t_1}^{t_2} a^2 dt, \quad (11.78)$$

with the stipulation that the *state of the system is identical at t_1 and t_2* . In the case of periodic motion, for instance, we must integrate over an integral number of full cycles.¹³ Now, the right-hand side of Eq. 11.78 can be integrated by parts:

$$\int_{t_1}^{t_2} a^2 dt = \int_{t_1}^{t_2} \left(\frac{d\mathbf{v}}{dt} \right) \cdot \left(\frac{d\mathbf{v}}{dt} \right) dt = \left(\mathbf{v} \cdot \frac{d\mathbf{v}}{dt} \right) \Big|_{t_1}^{t_2} - \int_{t_1}^{t_2} \frac{d^2\mathbf{v}}{dt^2} \cdot \mathbf{v} dt.$$

The boundary term drops out, since the velocities and accelerations are identical at t_1 and t_2 , so Eq. 11.78 can be written equivalently as

$$\int_{t_1}^{t_2} \left(\mathbf{F}_{\text{rad}} - \frac{\mu_0 q^2}{6\pi c} \dot{\mathbf{a}} \right) \cdot \mathbf{v} dt = 0. \quad (11.79)$$

Equation 11.79 will certainly be satisfied if

$$\boxed{\mathbf{F}_{\text{rad}} = \frac{\mu_0 q^2}{6\pi c} \dot{\mathbf{a}}.} \quad (11.80)$$

This is the **Abraham–Lorentz formula**¹⁴ for the radiation reaction force.

¹² Actually, while the total field is the sum of velocity and acceleration fields, $\mathbf{E} = \mathbf{E}_v + \mathbf{E}_a$, the *energy* is proportional to $E^2 = E_v^2 + 2\mathbf{E}_v \cdot \mathbf{E}_a + E_a^2$ and contains *three* terms: energy stored in the velocity fields alone (E_v^2), energy radiated away (E_a^2), and a *cross* term $\mathbf{E}_v \cdot \mathbf{E}_a$. For the sake of simplicity, I’m referring to the *combination* ($E_v^2 + 2\mathbf{E}_v \cdot \mathbf{E}_a$) as “energy stored in the nearby fields.” These terms go like $1/r^4$ and $1/r^3$, respectively, so neither one contributes to the radiation.

¹³ For *nonperiodic* motion the condition that the energy in the velocity fields be the same at t_1 and t_2 is more difficult to achieve. It is not enough that the instantaneous velocities and accelerations be equal, since the fields farther out depend on v and a at *earlier times*. In principle, then, v and a and *all higher derivatives* must be identical at t_1 and t_2 . In practice, since the velocity fields fall off rapidly with r , it is sufficient that v and a be the same over a brief interval prior to t_1 and t_2 .

¹⁴ This formula was first obtained by Lorentz in 1892 using the self-force method (Section 11.3.2); Planck pioneered the conservation-of-energy approach; Abraham

Of course, Eq. 11.79 doesn't *prove* Eq. 11.80. It says nothing whatever about the component of $\mathbf{F}_{\text{rad}}$ *perpendicular* to $\mathbf{v}$, and it only tells you the *time average* of the parallel component – the average, moreover, over very special time intervals. As we'll see in the next section, there are other reasons for believing in the Abraham–Lorentz formula, but for now, the best that can be said is that it represents the *simplest* form the radiation reaction force could take, consistent with conservation of energy.

The Abraham–Lorentz formula has disturbing implications, which are not entirely understood more than a century after the law was first proposed. For suppose a particle is subject to no *external* forces; then Newton's second law says

$$F_{\text{rad}} = \frac{\mu_0 q^2}{6\pi c} \dot{a} = ma,$$

from which it follows that

$$a(t) = a_0 e^{t/\tau}, \quad (11.81)$$

where

$$\tau \equiv \frac{\mu_0 q^2}{6\pi mc}. \quad (11.82)$$

(In the case of the electron, $\tau = 6 \times 10^{-24}$ s.) The acceleration spontaneously *increases* exponentially with time! This absurd conclusion can be avoided if we insist that $a_0 = 0$, but it turns out that the systematic exclusion of such **runaway solutions** has an even more unpleasant consequence: if you *do* apply an external force, the particle starts to respond *before the force acts*! (See Prob. 11.19.) This **acausal preacceleration** jumps the gun by only a short time τ ; nevertheless, it is (to my mind) unacceptable that the theory should countenance it at all.¹⁵

Example 11.4. Calculate the **radiation damping** of a charged particle attached to a spring of natural frequency ω_0 , driven at frequency ω .

Solution. The equation of motion is

$$m\ddot{x} = F_{\text{spring}} + F_{\text{rad}} + F_{\text{driving}} = -m\omega_0^2 x + m\tau \ddot{\dot{x}} + F_{\text{driving}}.$$

With the system oscillating at frequency ω ,

$$x(t) = x_0 \cos(\omega t + \delta),$$

so

$$\ddot{\dot{x}} = -\omega^2 \dot{x}.$$

polished both arguments, and generalized them to higher velocity; Dirac (1938) published a fully relativistic account. For the full history see K. T. McDonald, <http://kirkmcd.princeton.edu/examples/selfforce.pdf>.

¹⁵ These difficulties persist in the relativistic version of the Abraham–Lorentz equation, which can be derived by starting with Liénard's formula instead of Larmor's (Prob. 12.75). Perhaps they are telling us that there can be no such thing as a point charge in classical electrodynamics, or maybe they presage the onset of quantum mechanics. For guides to the literature, see P. Pearle's chapter in D. Teplitz, ed., *Electromagnetism: Paths to Research*, Plenum, New York (1982) and F. Rohrlich, *Am. J. Phys.* **65**, 1051 (1997).

Therefore

$$m\ddot{x} + m\gamma\dot{x} + m\omega_0^2x = F_{\text{driving}}, \quad (11.83)$$

and the damping factor γ is given by

$$\gamma = \omega^2\tau. \quad (11.84)$$

[When I wrote $F_{\text{damping}} = -\gamma mv$, back in Chapter 9 (Eq. 9.154), I assumed for simplicity that the damping was proportional to the velocity. We now know that *radiation* damping, at least, is proportional to $\ddot{v}$. But it hardly matters: for sinusoidal oscillations *any* even number of derivatives of v would do, since they're all proportional to v .]

Example 11.5. An electric dipole consists of a negative charge ($-q$) nailed down at the origin, and a positive charge ($+q$) held at rest on the z -axis ($z = d$). At time $t = 0$ the positive charge (mass m) is released, and accelerates toward the negative charge. Leaving aside the radiation reaction (which is a small perturbation) the force on q is

$$\mathbf{F} = -\frac{1}{4\pi\epsilon_0} \frac{q^2}{z^2} \hat{\mathbf{z}} = m\mathbf{a},$$

so its acceleration is

$$\mathbf{a} = \frac{\mathbf{F}}{m} = -\frac{q^2}{4\pi\epsilon_0 m} \frac{1}{z^2} \hat{\mathbf{z}}.$$

Differentiating with respect to t ,

$$\dot{\mathbf{a}} = -\frac{q^2}{4\pi\epsilon_0 m} \left(\frac{-2}{z^3} \right) \frac{dz}{dt} \hat{\mathbf{z}} = -\frac{2q^2}{4\pi\epsilon_0 m} \frac{v}{z^3} \hat{\mathbf{z}}$$

(note that $dz/dt = -v$, since the derivative is negative). The radiation reaction (Eq. 11.80) is therefore

$$\mathbf{F}_{\text{rad}} = \frac{\mu_0 q^2}{6\pi c} \dot{\mathbf{a}} = -\frac{\mu_0 q^4}{12\pi^2 \epsilon_0 m} \left(\frac{v}{c} \right) \frac{1}{z^3} \hat{\mathbf{z}}.$$

The details are of little interest – what's remarkable is the *sign*: it's in the same direction as $\mathbf{F}$ itself. This is not a brake, but an *accelerator*! At the same time that the system as a whole is losing energy in the form of radiation, q is actually *gaining* energy. (Of course, as it falls the particle gains kinetic energy and loses potential energy; absent the radiation reaction the *total* would be constant. We're talking about the *extra* energy associated with the radiation reaction.) Where is it coming from? It's coming from the fields, which are shedding energy in both directions.¹⁶ What does this tell us? Not much, I'm afraid, except to emphasize that the name “radiation reaction” can be seriously misleading.

¹⁶ Several authors have noted that the Poynting vector close to a decaying electric dipole points *inward*, whereas farther out it points *outward* (as it must, of course, to describe radiation). See H. G. Schantz, *Am. J. Phys.* **63**, 513 (1995), or K. T. McDonald, <http://kirkmcd.princeton.edu/examples/collapse.pdf>.

Problem 11.17.

- (a) A particle of charge q moves in a circle of radius R at a constant speed v . To sustain the motion, you must, of course, provide a centripetal force mv^2/R ; what *additional* force ($\mathbf{F}_e$) must you exert, in order to counteract the radiation reaction? [It's easiest to express the answer in terms of the instantaneous velocity $\mathbf{v}$.] What power (P_e) does this extra force deliver? Compare P_e with the power radiated (use the Larmor formula).
- (b) Repeat part (a) for a particle in simple harmonic motion with amplitude A and angular frequency ω : $\mathbf{w}(t) = A \cos(\omega t) \hat{\mathbf{z}}$. Explain the discrepancy.
- (c) Consider the case of a particle in free fall (constant acceleration g). What is the radiation reaction force? What is the power radiated? Comment on these results.

Problem 11.18. A point charge q , of mass m , is attached to a spring of constant k . At time $t = 0$ it is given a kick, so its initial energy is $U_0 = \frac{1}{2}mv_0^2$. Now it oscillates, gradually radiating away this energy.

- (a) Confirm that the total energy radiated is equal to U_0 . Assume the radiation damping is small, so you can write the equation of motion as

$$\ddot{x} + \gamma \dot{x} + \omega_0^2 x = 0,$$

and the solution as

$$x(t) = \frac{v_0}{\omega_0} e^{-\gamma t/2} \sin(\omega_0 t),$$

with $\omega_0 \equiv \sqrt{k/m}$, $\gamma = \omega_0^2 \tau$, and $\gamma \ll \omega_0$ (drop γ^2 in comparison to ω_0^2 , and when you average over a complete cycle, ignore the change in $e^{-\gamma t}$).

- (b) Suppose now we have *two* such oscillators, and we start them off with identical kicks. Regardless of their relative positions and orientations, the total energy radiated is $2U_0$. But what if they are right on top of each other, so it's equivalent to a *single* oscillator with twice the charge; the Larmor formula says that the power radiated is *four* times as great, suggesting that the total will be $4U_0$. Find the error in this reasoning, and show that the total is actually $2U_0$, as it should be.¹⁷

- ! **Problem 11.19.** With the inclusion of the radiation reaction force (Eq. 11.80, Newton's second law for a charged particle becomes

$$a = \tau \dot{a} + \frac{F}{m},$$

where F is the external force acting on the particle.

- (a) In contrast to the case of an *uncharged* particle ($a = F/m$), acceleration (like position and velocity) must now be a *continuous* function of time, even if the force changes abruptly. (Physically, the radiation reaction damps out any rapid change in a .) *Prove* that a is

¹⁷ For a more sophisticated version of this paradox, see P. R. Berman, *Am. J. Phys.* **78**, 1323 (2010).

continuous at any time t , by integrating the equation of motion from $(t - \epsilon)$ to $(t + \epsilon)$ and taking the limit $\epsilon \rightarrow 0$.

- (b) A particle is subjected to a constant force F , beginning at time $t = 0$ and lasting until time T . Find the most general solution $a(t)$ to the equation of motion in each of the following three periods: (i) $t < 0$, (ii) $0 < t < T$, (iii) $t > T$.
- (c) Impose the continuity condition (a) at $t = 0$ and $t = T$. Show that you can *either* eliminate the runaway in region (iii) *or* avoid preacceleration in region (i), but not both.
- (d) If you choose to eliminate the runaway, what is the acceleration as a function of time, in each interval? How about the velocity? (The latter must, of course, be continuous at $t = 0$ and $t = T$.) Assume the particle was originally at rest: $v(-\infty) = 0$.
- (e) Plot $a(t)$ and $v(t)$, both for an uncharged particle and for a (nonrunaway) charged particle, subject to this force.

11.3.2 The Self-Force on a Charged Particle

In the last section, I derived the Abraham–Lorentz formula for the radiation reaction, using conservation of energy. I made no attempt to identify the actual *mechanism* responsible for this force, except to point out that it must be a recoil effect of the particle’s own fields acting back on the charge. Unfortunately, the fields of a point charge blow up right at the particle, so it’s hard to see how one might calculate the force they exert.¹⁸ Let’s avoid this problem by considering an *extended* charge distribution, for which the field is finite everywhere; at the end, we’ll take the limit as the size of the charge goes to zero. In general, as I noted in Section 8.2.1, the electromagnetic force of one part (A) on another part (B) is *not* equal and opposite to the force of B on A (Fig. 11.16). If the distribution is divided up into infinitesimal chunks, and the imbalances are added up for all such pairs, the result is a *net force of the charge* on itself. It is this **self-force**, resulting from the breakdown of Newton’s third law within the structure of the particle, that accounts for the radiation reaction.

Lorentz originally calculated the electromagnetic self-force using a spherical charge distribution, which seems reasonable but makes the mathematics rather cumbersome.¹⁹ Because I am only trying to elucidate the *mechanism* involved, I shall use a less realistic model: a “dumbbell” in which the total charge q is divided into two halves separated by a fixed distance d (Fig. 11.17). This is the simplest possible arrangement of the charge that permits the essential mechanism (imbalance of internal electromagnetic forces) to function. Never mind that it’s an unlikely model for an elementary particle: in the point limit ($d \rightarrow 0$) *any* model must yield

¹⁸ It can be done by a suitable averaging of the field, but it’s not easy. See T. H. Boyer, *Am. J. Phys.* **40**, 1843 (1972), and references cited there.

¹⁹ See J. D. Jackson, *Classical Electrodynamics*, 3rd ed., John Wiley, New York (1999), Sect. 16.3.

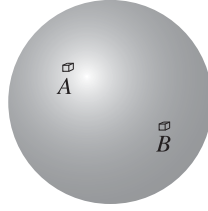


Fig. 11.16

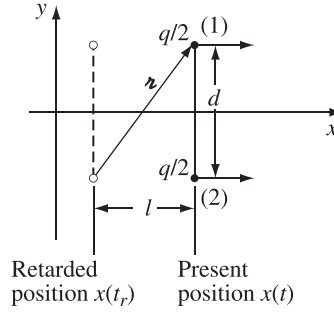


Fig. 11.17

the Abraham–Lorentz formula, to the extent that conservation of energy alone dictates that answer.

Let's assume the dumbbell moves in the x direction, and is (instantaneously) at rest at the retarded time. The electric field at (1) due to (2) is

$$\mathbf{E}_1 = \frac{(q/2)}{4\pi\epsilon_0} \frac{r}{(\mathbf{r} \cdot \mathbf{u})^3} [(c^2 + \mathbf{r} \cdot \mathbf{a})\mathbf{u} - (\mathbf{r} \cdot \mathbf{u})\mathbf{a}] \quad (11.85)$$

(Eq. 10.72), where

$$\mathbf{u} = c\hat{\mathbf{r}} \quad \text{and} \quad \mathbf{r} = l\hat{\mathbf{x}} + d\hat{\mathbf{y}}, \quad (11.86)$$

so

$$\mathbf{r} \cdot \mathbf{u} = cr, \quad \mathbf{r} \cdot \mathbf{a} = la, \quad \text{and} \quad r = \sqrt{l^2 + d^2}. \quad (11.87)$$

Actually, we're only interested in the x component of $\mathbf{E}_1$, since the y components will cancel when we add the forces on the two ends (for the same reason, we don't need to worry about the magnetic forces). Now

$$u_x = \frac{cl}{r}, \quad (11.88)$$

and hence

$$E_{1x} = \frac{q}{8\pi\epsilon_0 c^2} \frac{(lc^2 - ad^2)}{(l^2 + d^2)^{3/2}}. \quad (11.89)$$

By symmetry, $E_{2x} = E_{1x}$, so the net force on the dumbbell is

$$\mathbf{F}_{\text{self}} = \frac{q}{2}(\mathbf{E}_1 + \mathbf{E}_2) = \frac{q^2}{8\pi\epsilon_0 c^2} \frac{(lc^2 - ad^2)}{(l^2 + d^2)^{3/2}} \hat{\mathbf{x}}. \quad (11.90)$$

So far everything is exact. The idea now is to expand in powers of d ; when the size of the particle goes to zero, all *positive* powers will disappear. Using Taylor's theorem,

$$x(t) = x(t_r) + \dot{x}(t_r)(t - t_r) + \frac{1}{2}\ddot{x}(t_r)(t - t_r)^2 + \frac{1}{3!}\dddot{x}(t_r)(t - t_r)^3 + \cdots,$$

we have

$$l = x(t) - x(t_r) = \frac{1}{2}aT^2 + \frac{1}{6}\dot{a}T^3 + \cdots, \quad (11.91)$$

where $T \equiv t - t_r$, for short. Now T is determined by the retarded time condition

$$(cT)^2 = l^2 + d^2, \quad (11.92)$$

so

$$d = \sqrt{(cT)^2 - l^2} = cT \sqrt{1 - \left(\frac{aT}{2c} + \frac{\dot{a}T^2}{6c} + \dots \right)^2} = cT - \frac{a^2}{8c} T^3 + () T^4 + \dots .$$

This equation tells us d , in terms of T ; we need to “solve” it for T as a function of d . There’s a systematic procedure for doing this, known as **reversion of series**,²⁰ but we can get the first couple of terms more informally as follows: ignoring all higher powers of T ,

$$d \approx cT \Rightarrow T \approx \frac{d}{c};$$

using this as an approximation for the cubic term,

$$d \approx cT - \frac{a^2}{8c} \frac{d^3}{c^3} \Rightarrow T \approx \frac{d}{c} + \frac{a^2 d^3}{8c^5},$$

and so on. Thus

$$T = \frac{1}{c} d + \frac{a^2}{8c^5} d^3 + () d^4 + \dots . \quad (11.93)$$

Returning to Eq. 11.91, we construct the power series for l in terms of d :

$$l = \frac{a}{2c^2} d^2 + \frac{\dot{a}}{6c^3} d^3 + () d^4 + \dots . \quad (11.94)$$

Putting this into Eq. 11.90, I conclude that

$$\mathbf{F}_{\text{self}} = \frac{q^2}{4\pi\epsilon_0} \left[-\frac{a}{4c^2 d} + \frac{\dot{a}}{12c^3} + () d + \dots \right] \hat{\mathbf{x}}. \quad (11.95)$$

Here a and $\dot{a}$ are evaluated at the *retarded* time (t_r), but it’s easy to rewrite the result in terms of the *present* time t :

$$a(t_r) = a(t) + \dot{a}(t)(t_r - t) + \dots = a(t) - \dot{a}(t)T + \dots = a(t) - \dot{a}(t) \frac{d}{c} + \dots ,$$

and it follows that

$$\mathbf{F}_{\text{self}} = \frac{q^2}{4\pi\epsilon_0} \left[-\frac{a(t)}{4c^2 d} + \frac{\dot{a}(t)}{3c^3} + () d + \dots \right] \hat{\mathbf{x}}. \quad (11.96)$$

The first term on the right is proportional to the acceleration of the charge; if we pull it over to the other side of Newton’s second law, it simply adds to the dumbbell’s mass. In effect, the total inertia of the charged dumbbell is

$$m = 2m_0 + \frac{1}{4\pi\epsilon_0} \frac{q^2}{4dc^2}, \quad (11.97)$$

²⁰ See, for example, the *CRC Standard Mathematical Tables and Formulas*, 32nd ed., CRC Press, Boca Raton (2011).

where m_0 is the mass of either end alone. In the context of special relativity, it is not surprising that the electrical repulsion of the charges should enhance the mass of the dumbbell. The potential energy of this configuration (in the static case) is

$$\frac{1}{4\pi\epsilon_0} \frac{(q/2)^2}{d}, \quad (11.98)$$

and according to Einstein's formula $E = mc^2$, this energy contributes to the inertia of the object.²¹

The second term in Eq. 11.96 is the radiation reaction:

$$F_{\text{rad}}^{\text{int}} = \frac{\mu_0 q^2 \dot{a}}{12\pi c}. \quad (11.99)$$

It alone (apart from the mass correction²²) survives in the “point dumbbell” limit $d \rightarrow 0$. Unfortunately, it differs from the Abraham–Lorentz formula by a factor of 2. But then, this is only the self-force associated with the *interaction* between 1 and 2 – hence the superscript “int.” There remains the force of *each end on itself*. When the latter is included (see Prob. 11.20), the result is

$$F_{\text{rad}} = \frac{\mu_0 q^2 \dot{a}}{6\pi c}, \quad (11.100)$$

reproducing the Abraham–Lorentz formula exactly. *Conclusion: The radiation reaction is due to the force of the charge on itself* – or, more elaborately, the net force exerted by the fields generated by different parts of the charge distribution acting on one another.

Problem 11.20. Deduce Eq. 11.100 from Eq. 11.99. Here are three methods:

- Use the Abraham–Lorentz formula to determine the radiation reaction on each end of the dumbbell; add this to the interaction term (Eq. 11.99).
- Method (a) has the defect that it *uses* the Abraham–Lorentz formula – the very thing that we were trying to *derive*. To avoid this, let $F(q)$ be the total d -independent part of the self-force on a charge q . Then

$$F(q) = F^{\text{int}}(q) + 2F(q/2),$$

²¹ The fact that the *numbers* work out perfectly is a lucky feature of this configuration. If you do the same calculation for the dumbbell in *longitudinal* motion, the mass correction is only half of what it “should” be (there’s a 2, instead of a 4, in Eq. 11.97), and for a sphere it’s off by a factor of 3/4. This notorious paradox has been the subject of much debate over the years. See D. J. Griffiths and R. E. Owen, *Am. J. Phys.* **51**, 1120 (1983).

²² Of course, the limit $d \rightarrow 0$ has an embarrassing effect on the mass term. In a sense, it doesn’t matter, since only the *total* mass m is observable; maybe m_0 somehow has a compensating (negative!) infinity, so that m comes out finite. This awkward problem persists in quantum electrodynamics, where it is “swept under the rug” in a procedure known as **mass renormalization**.

where F^{int} is the interaction part (Eq. 11.99), and $F(q/2)$ is the self-force on each end. Now, $F(q)$ must be proportional to q^2 , since the field is proportional to q and the force is $q\mathbf{E}$. So $F(q/2) = (1/4)F(q)$. Take it from there.

- (c) Smear out the charge along a strip of length L oriented perpendicular to the motion (the charge density, then, is $\lambda = q/L$); find the cumulative interaction force for all pairs of segments, using Eq. 11.99 (with the correspondence $q/2 \rightarrow \lambda dy_1$, at one end and $q/2 \rightarrow \lambda dy_2$ at the other). Make sure you don't count the same pair twice.

! **Problem 11.21.**²³ An electric dipole rotates at constant angular velocity ω in the xy plane. (The charges, $\pm q$, are at $\mathbf{r}_{\pm} = \pm R[\cos(\omega t)\hat{\mathbf{x}} + \sin(\omega t)\hat{\mathbf{y}}]$; the magnitude of the dipole moment is $p = 2qR$.)

- (a) Find the interaction term in the self-torque (analogous to Eq. 11.99). Assume the motion is nonrelativistic ($\omega R \ll c$).
- (b) Use the method of Prob. 11.20a to obtain the *total* radiation reaction torque on this system. [Answer: $-(\mu_0 p^2 \omega^3) \hat{\mathbf{z}}/6\pi c$.]
- (c) Check that this result is consistent with the power radiated (Eq. 11.60).

More Problems on Chapter 11

Problem 11.22. A particle of mass m and charge q is attached to a spring with force constant k , hanging from the ceiling (Fig. 11.18). Its equilibrium position is a distance h above the floor. It is pulled down a distance d below equilibrium and released, at time $t = 0$.

- (a) Under the usual assumptions ($d \ll \lambda \ll h$), calculate the intensity of the radiation hitting the floor, as a function of the distance R from the point directly below q . [Note: The intensity here is the average power per unit area of the floor.] At what R is the radiation most intense? Neglect the radiative damping of the oscillator. [Answer: $\mu_0 q^2 d^2 \omega^4 R^2 h / 32 \pi^2 c (R^2 + h^2)^{5/2}$.]

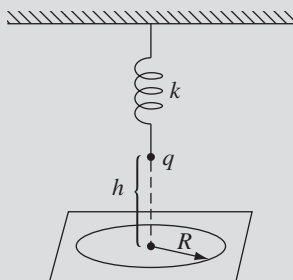


Fig. 11.18

²³ For related problems, see D. R. Stump and G. L. Pollack, *Am. J. Phys.* **65**, 81 (1997); D. Griffiths and E. Szeto, *Am. J. Phys.* **46**, 244 (1978).

- (b) As a check on your formula, assume the floor is of infinite extent, and calculate the average energy per unit time striking the entire floor. Is it what you'd expect?
- (c) Because it is losing energy in the form of radiation, the amplitude of the oscillation will gradually decrease. After what time τ has the amplitude been reduced to d/e ? (Assume the fraction of the total energy lost in one cycle is very small.)

Problem 11.23. A radio tower rises to height h above flat horizontal ground. At the top is a magnetic dipole antenna, of radius b , with its axis vertical. FM station KRUD broadcasts from this antenna at (angular) frequency ω , with a total radiated power P (that's averaged, of course, over a full cycle). Neighbors have complained about problems they attribute to excessive radiation from the tower – interference with their stereo systems, mechanical garage doors opening and closing mysteriously, and a variety of suspicious medical problems. But the city engineer who measured the radiation level at the base of the tower found it to be well below the accepted standard. You have been hired by the Neighborhood Association to assess the engineer's report.

- (a) In terms of the variables given (not all of which may be relevant), find the formula for the intensity of the radiation at ground level, a distance R from the base of the tower. You may assume that $b \ll c/\omega \ll h$. [Note: We are interested only in the *magnitude* of the radiation, not in its *direction* – when measurements are taken, the detector will be aimed directly at the antenna.]
- (b) How far from the base of the tower *should* the engineer have made the measurement? What is the formula for the intensity at this location?
- (c) KRUD's actual power output is 35 kW, its frequency is 90 MHz, the antenna's radius is 6 cm, and the height of the tower is 200 m. The city's radio-emission limit is $200 \mu\text{W}/\text{cm}^2$. Is KRUD in compliance?

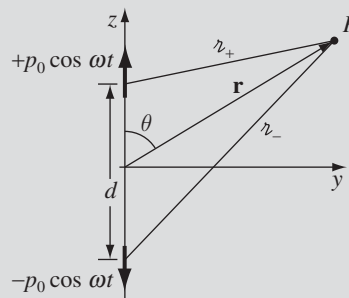


Fig. 11.19

- ! **Problem 11.24.** As a model for electric quadrupole radiation, consider two oppositely oriented oscillating electric dipoles, separated by a distance d , as shown in Fig. 11.19. Use the results of Section 11.1.2 for the potentials of each dipole, but note that they are *not* located at the origin. Keeping only the terms of first order in d :

- (a) find the scalar and vector potentials;
- (b) find the electric and magnetic fields;

- (c) find the Poynting vector and the power radiated, then sketch the intensity profile as a function of θ .

Problem 11.25. The *geographic* poles of the Earth are defined by its axis of rotation; the *magnetic* poles are off (from this axis) by about 11° (and they wander a bit). The north pole of a magnetic compass points (roughly) northward, so the Earth's magnetic *south* pole is actually pretty close to its geographic *north* pole. (Why is this so complicated?) Relative to the fixed axis of rotation, therefore, the magnetic dipole moment of the Earth is changing with time, and the Earth must be giving off magnetic dipole radiation.

- (a) Find the formula for the total power radiated, in terms of the following parameters: ψ (the angle between the geographic and magnetic poles), M (the magnitude of the Earth's magnetic dipole moment), and ω (the angular velocity of rotation of the Earth). [Hint: Refer to Prob. 11.4 or Prob. 11.11.]
- (b) Using the fact that the Earth's magnetic field is about half a gauss at the equator, estimate the magnetic dipole moment M of the Earth.
- (c) Find the power radiated. [Answer: $4 \times 10^{-5} \text{ W}$.]
- (d) Pulsars are thought to be rotating neutron stars, with a typical radius of 10 km, a rotational period of 10^{-3} s , and a surface magnetic field of 10^8 T . What sort of radiated power would you expect from such a star?²⁴ [Answer: $2 \times 10^{36} \text{ W}$.]

Problem 11.26. An ideal electric dipole is situated at the origin; its dipole moment points in the $\hat{\mathbf{z}}$ direction, and is quadratic in time:

$$\mathbf{p}(t) = \frac{1}{2}\ddot{p}_0 t^2 \hat{\mathbf{z}} \quad (-\infty < t < \infty),$$

where $\ddot{p}_0$ is a constant.

- (a) Use the method of Section 11.1.2 to determine the (exact) electric and magnetic fields, for all $r > 0$ (there's also a delta-function term at the origin, but we're not concerned with that). [Partial answer: $V = (\mu_0 \ddot{p}_0 / 8\pi) \cos \theta [(ct/r)^2 - 1]$, $\mathbf{A} = (\mu_0 \ddot{p}_0 / 4\pi c) [(ct/r) - 1] \hat{\mathbf{z}}$.]
- (b) Calculate the power, $P(r, t)$, passing through a sphere of radius r . [Answer: $(\ddot{p}_0^2 / 12\pi\epsilon_0 r^3) t[t^2 + (r/c)^2]$.]
- (c) Find the total power radiated (Eq. 11.2), and check that your answer is consistent with Eq. 11.60.²⁵

- ! **Problem 11.27.** In Section 11.2.1 we calculated the *energy* per unit time radiated by a (nonrelativistic) point charge – the Larmor formula.

²⁴ J. P. Ostriker and J. E. Gunn, *Astrophys. J.* **157**, 1395 (1969).

²⁵ Notice that $\mathbf{B}(r, t)$ goes like $1/r^2$, and one might therefore assume that this configuration does not radiate. However, it is not $\mathbf{B}(r, t)$ we require (for Eq. 11.2), but rather $\mathbf{B}(r, t_0 + r/c)$ – we track the fields as they propagate out to infinity – and $\mathbf{B}(r, t_0 + r/c)$ has a term that goes like $1/r$.

- (a) In the same spirit, calculate the *momentum* per unit time radiated. [Answer: $(\mu_0 q^2 / 6\pi c^3) a^2 \mathbf{v}$.]
 (b) Similarly, calculate the *angular momentum* per unit time radiated. [Answer: $(\mu_0 q^2 / 6\pi c)(\mathbf{v} \times \mathbf{a})$.]

Problem 11.28. Suppose the (electrically neutral) yz plane carries a time-dependent but uniform surface current $K(t)\hat{\mathbf{z}}$.

- (a) Find the electric and magnetic fields at a height x above the plane if
 (i) a constant current is turned on at $t = 0$:

$$K(t) = \begin{cases} 0, & t \leq 0, \\ K_0, & t > 0. \end{cases}$$

- (ii) a linearly increasing current is turned on at $t = 0$:

$$K(t) = \begin{cases} 0, & t \leq 0, \\ \alpha t, & t > 0. \end{cases}$$

- (b) Show that the retarded vector potential can be written in the form

$$\mathbf{A}(x, t) = \frac{\mu_0 c}{2} \hat{\mathbf{z}} \int_0^\infty K\left(t - \frac{x}{c} - u\right) du,$$

and from this determine $\mathbf{E}$ and $\mathbf{B}$.

- (c) Show that the total power radiated per unit area of surface is

$$\frac{\mu_0 c}{2} [K(t)]^2.$$

Explain what you mean by “radiation,” in this case, given that the source is not localized.²⁶

Problem 11.29. Use the duality transformation (Prob. 7.68) to construct the electric and magnetic fields of a magnetic monopole q_m in arbitrary motion, and find the “Larmor formula” for the power radiated.²⁷

Problem 11.30. Assuming you exclude the runaway solution in Prob. 11.19, calculate:

- (a) the work done by the external force;
 (b) the final kinetic energy (assume the initial kinetic energy was zero);
 (c) the total energy radiated.
 (d) Check that energy is conserved in this process.²⁸

²⁶ For discussion and related problems, see B. R. Holstein, *Am. J. Phys.* **63**, 217 (1995); T. A. Abbott and D. J. Griffiths, *Am. J. Phys.* **53**, 1203 (1985).

²⁷ For related applications, see J. A. Heras, *Am. J. Phys.* **63**, 242 (1995).

²⁸ Problems 11.30 and 11.31 were suggested by G. L. Pollack.

Problem 11.31.

- (a) Repeat Prob. 11.19, but this time let the external force be a Dirac delta function: $F(t) = k\delta(t)$ (for some constant k).²⁹ [Note that the acceleration is now *discontinuous* at $t = 0$ (though the *velocity* must still be continuous); use the method of Prob. 11.19a to show that $\Delta a = -k/m\tau$. In this problem there are only *two* intervals to consider: (i) $t < 0$ and (ii) $t > 0$.]
- (b) As in Prob. 11.30, check that energy is conserved in this process.

- ! **Problem 11.32.** A charged particle, traveling in from $-\infty$ along the x -axis, encounters a rectangular potential energy barrier

$$U(x) = \begin{cases} U_0 & \text{if } 0 < x < L, \\ 0 & \text{otherwise.} \end{cases}$$

Show that, because of the radiation reaction, it is possible for the particle to **tunnel** through the barrier – that is, even if the incident kinetic energy is less than U_0 , the particle can pass through.³⁰ [Hint: Your task is to solve the equation

$$a = \tau \dot{a} + \frac{F}{m},$$

subject to the force

$$F(x) = U_0[-\delta(x) + \delta(x - L)].$$

Refer to Probs. 11.19 and 11.31, but notice that this time the force is a specified function of x , not t . There are three regions to consider: (i) $x < 0$, (ii) $0 < x < L$, (iii) $x > L$. Find the general solution for $a(t)$, $v(t)$, and $x(t)$ in each region, exclude the runaway in region (iii), and impose the appropriate boundary conditions at $x = 0$ and $x = L$. Show that the final velocity (v_f) is related to the time T spent traversing the barrier by the equation

$$L = v_f T - \frac{U_0}{mv_f} (\tau e^{-T/\tau} + T - \tau),$$

and the initial velocity (at $x = -\infty$) is

$$v_i = v_f - \frac{U_0}{mv_f} \left[1 - \frac{1}{1 + \frac{U_0}{mv_f^2} (e^{-T/\tau} - 1)} \right].$$

To simplify these results (since all we're looking for is a specific example), suppose the final kinetic energy is half the barrier height. Show that in this case

$$v_i = \frac{v_f}{1 - (L/v_f\tau)}.$$

In particular, if you choose $L = v_f\tau/4$, then $v_i = (4/3)v_f$, the initial kinetic energy is $(8/9)U_0$, and the particle makes it through, even though it didn't have sufficient energy to get over the barrier!]

²⁹ This example was first analyzed by P. A. M. Dirac, *Proc. R. Soc. A* **167**, 148 (1938).

³⁰ F. Denef et al., *Phys. Rev. E* **56**, 3624 (1997).

! **Problem 11.33.**

- (a) Find the radiation reaction force on a particle moving with arbitrary velocity in a straight line, by reconstructing the argument in Section 11.3.2 *without* assuming $v(t_r) = 0$. [Answer: $(\mu_0 q^2 \gamma^4 / 6\pi c)(\dot{a} + 3\gamma^2 a^2 v/c^2)$.]
- (b) Show that this result is consistent (in the sense of Eq. 11.78) with the power radiated by such a particle (Eq. 11.75).

Problem 11.34.

- (a) Does a particle in hyperbolic motion (Eq. 10.52) radiate? (Use the exact formula (Eq. 11.75) to calculate the power radiated.)
- (b) Does a particle in hyperbolic motion experience a radiation reaction? (Use the exact formula (Prob. 11.33) to determine the reaction force.)

[Comment: These famous questions carry important implications for the **principle of equivalence**.³¹]

Problem 11.35. Use the result of Prob. 10.35 to determine the power radiated by an ideal electric dipole, $\mathbf{p}(t)$, at the origin. Check that your answer is consistent with Eq. 11.22, in the case of sinusoidal time dependence, and with Prob. 11.26, in the case of quadratic time dependence.

Problem 11.36. An infinite plane (might as well make it the yz plane) is electrically neutral but carries a sinusoidal surface current

$$\mathbf{K}(t) = K_0 \cos(\omega t) \hat{\mathbf{z}}.$$

- (a) By whatever means you can think of (short of plagiarism) deduce the electric and magnetic fields it produces. (If you use guesswork or divine inspiration, be sure to check that your results satisfy Maxwell's equations and the appropriate boundary conditions.)
- (b) Construct the Poynting vector and its time average. By integrating the latter over a "pillbox" straddling the current, determine the (average) power radiated, per unit area.
- (c) Check that your answer to (b) is consistent with the power delivered to the surface by whoever is driving the current.

³¹ T. Fulton and F. Rohrlich, *Ann. Phys.* **9**, 499 (1960); J. Cohn, *Am. J. Phys.* **46**, 225 (1978); R. Peierls, *Surprises in Theoretical Physics*, Princeton University Press, Princeton, NJ (1979), Chap. 8; the article by P. Pearle in D. Teplitz, ed., *Electromagnetism: Paths to Research*, Plenum Press, New York (1982); C. de Almeida and A. Saa, *Am. J. Phys.* **74**, 154 (2006).